

Experiment No.4

Aim: To study and Implement Infrastructure as a Service using AWS/Microsoft Azure.

Objective:

- Gain a theoretical understanding of Infrastructure as a Service (IaaS) and its core concepts.
- Apply practical knowledge to set up and configure virtualized infrastructure components on AWS/Azure.

Outcomes:

- Students will demonstrate proficiency in utilizing AWS IaaS services.
- Students will apply theoretical knowledge to implement a functional cloud infrastructure on AWS.

Theory:

Infrastructure as a Service (IaaS) is a cloud computing model that provides virtualized computing resources over the internet. AWS, a leading cloud service provider, offers a range of IaaS services, allowing users to provision and manage virtual machines, storage, and networking components on a pay-as-you-go basis. Key AWS services for IaaS include Amazon EC2 for virtual servers, Amazon S3 for scalable storage, and Amazon VPC for networking.

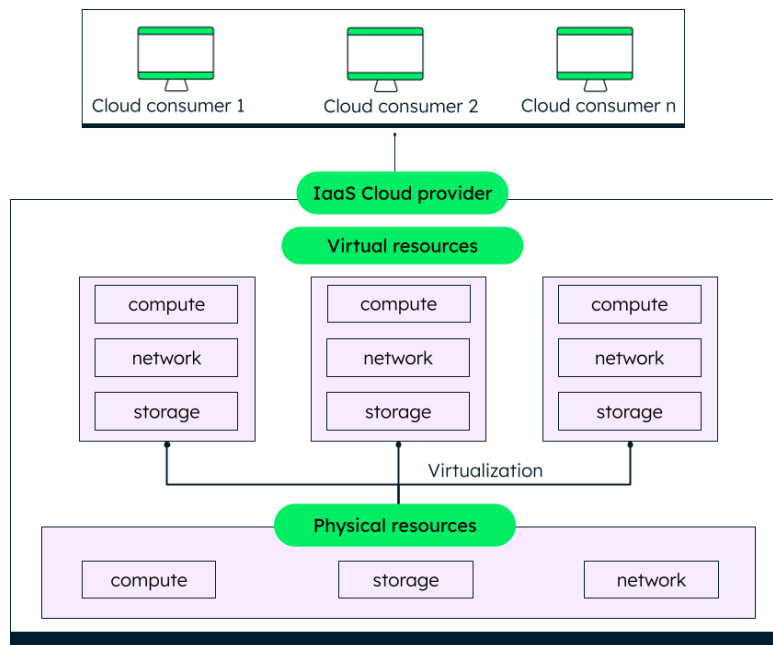


Figure 5 Basic components of IaaS, namely, compute, network and storage

Understanding the Building Blocks:

1. **Physical Infrastructure:** The foundation of IaaS lies in vast physical data centers brimming with servers, storage arrays, and networking equipment. These resources are pooled and virtualized, creating a ready-to-use pool of computing power.
2. **Hypervisor:** This software layer sits atop the physical hardware, slicing it into smaller, virtual partitions. Each partition acts as a virtual machine (VM), capable of running its own operating system and applications, independent of the underlying hardware.
3. **Management Platform:** This web-based portal serves as your control center for IaaS. It allows you to provision virtual resources, configure networks, manage storage, and monitor performance.
4. **API Access:** For programmatic control, IaaS providers offer Application Programming Interfaces (APIs). These APIs enable automation and seamless integration of IaaS resources with your existing IT infrastructure.

The IaaS Advantage:

- **Agility and Scalability:** On-demand provisioning lets you scale your infrastructure up or down instantly, adapting to changing workloads and avoiding over-provisioning.
- **Cost-Effectiveness:** Pay only for the resources you use, eliminating the need for upfront investments in physical hardware.
- **Flexibility:** Choose from a wide range of pre-configured VM sizes and operating systems, or customize your own configuration.
- **High Availability:** IaaS providers ensure redundancy and disaster recovery measures, keeping your applications and data accessible even in case of hardware failures.
- **Global Reach:** Access your infrastructure from anywhere, anytime, and leverage geographically distributed data centers for optimal performance.

Real-World Applications:

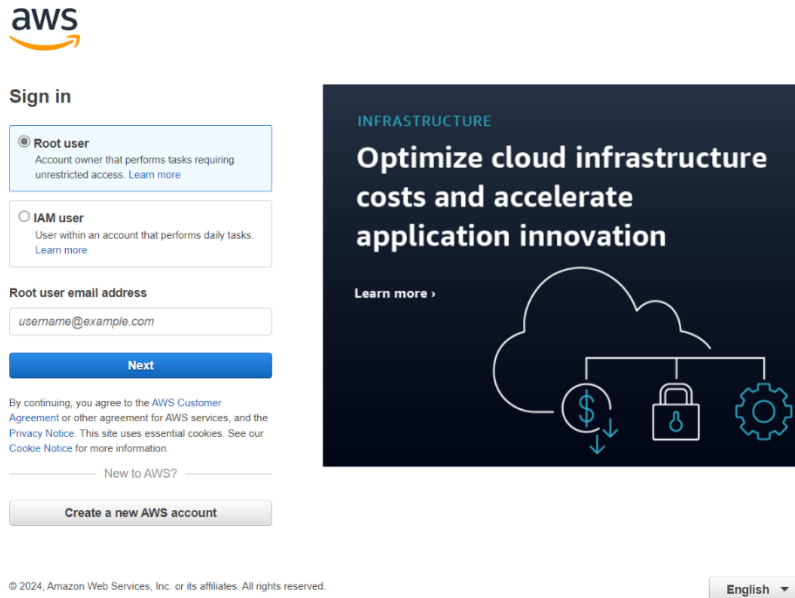
IaaS empowers diverse use cases across various industries:

- **Web hosting:** Host websites and applications with on-demand scalability and global reach.
- **Development and testing:** Create temporary environments for software development and testing without impacting production systems.
- **Big data and analytics:** Run data analytics applications on scalable clusters of virtual machines.
- **Disaster recovery:** Ensure business continuity by replicating mission-critical applications to a secondary IaaS environment.

Stepwise Procedure to Implement IaaS using AWS:

Step 1: Sign into AWS Console

- Log in to the AWS Management Console using your AWS account credentials.



aws

Sign in

☒ **Root user**
Account owner that performs tasks requiring unrestricted access. [Learn more](#)

☐ **IAM user**
User within an account that performs daily tasks. [Learn more](#)

Root user email address
username@example.com

Next

By continuing, you agree to the [AWS Customer Agreement](#) or other agreement for AWS services, and the [Privacy Notice](#). This site uses essential cookies. See our [Cookie Notice](#) for more information.

[New to AWS?](#)

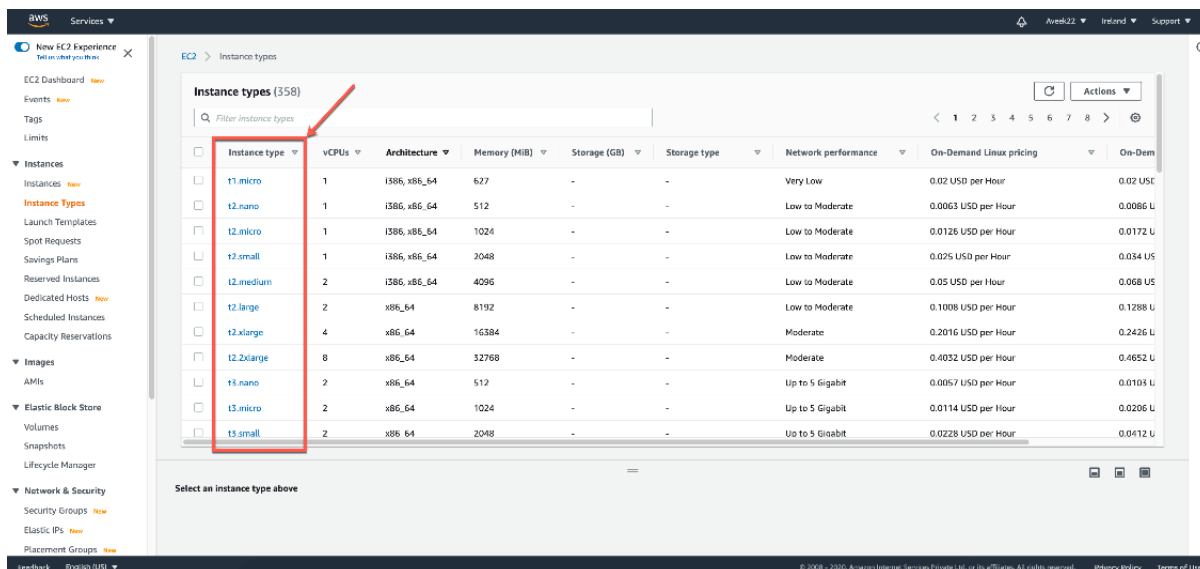
Create a new AWS account

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English

Step 2: Launch an EC2 Instance

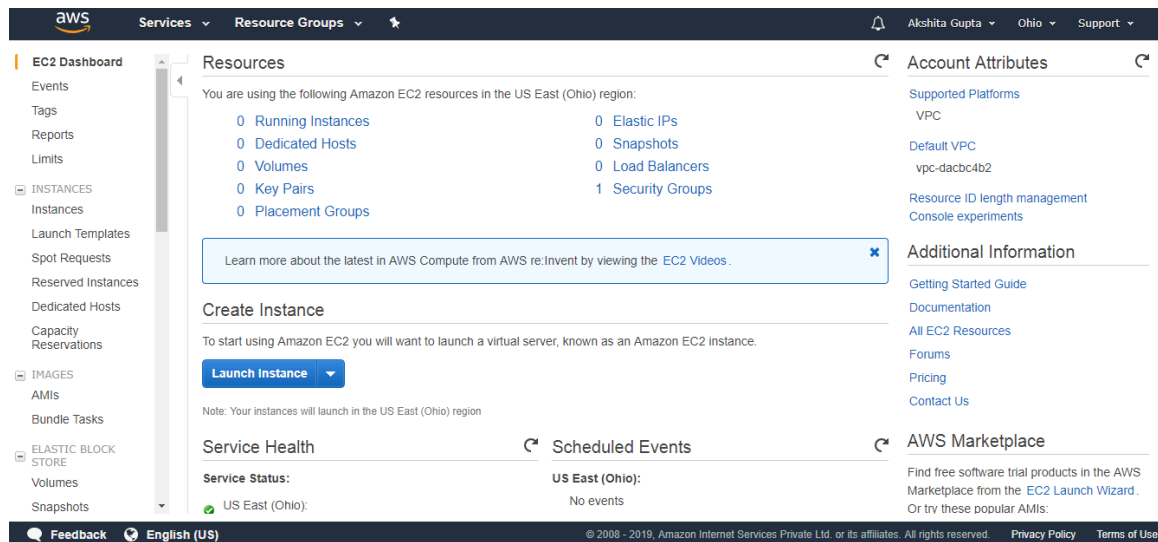
- Navigate to the EC2 service.
- Click "Launch Instance" to create a virtual server (EC2 instance).
- Choose an Amazon Machine Image (AMI), configure instance details, add storage, and configure security groups.



Instance type	vCPUs	Architecture	Memory (MiB)	Storage (GB)	Storage type	Network performance	On-Demand Linux pricing	On-Dem
☐ t1.micro	1	i386, x86_64	627	-	-	Very Low	0.02 USD per Hour	0.02 USD
☐ t2.nano	1	i386, x86_64	512	-	-	Low to Moderate	0.0063 USD per Hour	0.0086 U
☐ t2.micro	1	i386, x86_64	1024	-	-	Low to Moderate	0.0126 USD per Hour	0.0172 U
☐ t2.small	1	i386, x86_64	2048	-	-	Low to Moderate	0.025 USD per Hour	0.034 US
☐ t2.medium	2	i386, x86_64	4096	-	-	Low to Moderate	0.05 USD per Hour	0.068 U
☐ t2.large	2	x86_64	8192	-	-	Low to Moderate	0.1008 USD per Hour	0.1288 U
☐ t2.xlarge	4	x86_64	16384	-	-	Moderate	0.2016 USD per Hour	0.2436 U
☐ t2.2xlarge	8	x86_64	32768	-	-	Moderate	0.4032 USD per Hour	0.4872 U
☐ t3.nano	2	x86_64	512	-	-	Up to 5 Gigabit	0.0057 USD per Hour	0.0103 U
☐ t3.micro	2	x86_64	1024	-	-	Up to 5 Gigabit	0.0114 USD per Hour	0.0206 U
☐ t3.small	2	x86_64	2048	-	-	Up to 5 Gigabit	0.0228 USD per Hour	0.0412 U

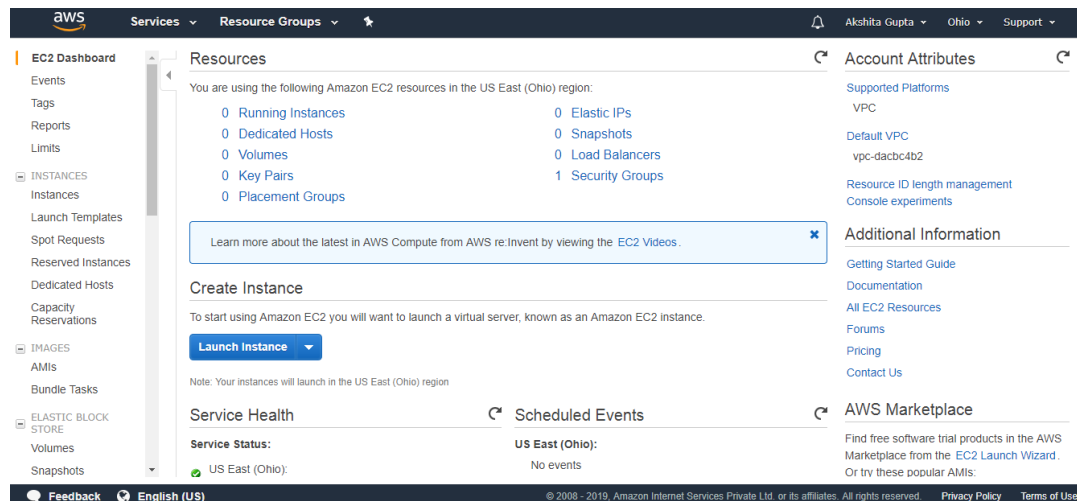
Step 3: Create an Amazon S3 Bucket

- Go to the S3 service in the AWS Console.
- Click "Create Bucket" to set up scalable object storage.
- Configure bucket settings, set permissions, and create the S3 bucket.



Step 4: Set Up Amazon VPC

- Access the VPC service.
- Create a new Virtual Private Cloud (VPC), configure subnets, and set up routing tables.



Step 5: Connect EC2 Instance to VPC

- Modify the EC2 instance settings to connect it to the created VPC.
- Update security groups and networking configurations.

```
[ec2-user@ip-172-31-46-94 ~]$  
[ec2-user@ip-172-31-46-94 ~]$  
[ec2-user@ip-172-31-46-94 ~]$  
[ec2-user@ip-172-31-46-94 ~]$  
[ec2-user@ip-172-31-46-94 ~]$ ssh ec2-user@172.31.101.108  
The authenticity of host '172.31.101.108 (172.31.101.108)' can't be established  
.  
ECDSA key fingerprint is SHA256:yWcSCv5zhzq1oiqUcIy0yhXMKiH2D2w9SfH0hA6BbOI.  
ECDSA key fingerprint is MD5:2a:6b:ca:b5:54:ed:d2:c7:42:a0:8f:da:43:99:25:16.  
Are you sure you want to continue connecting (yes/no)? yes  
Warning: Permanently added '172.31.101.108' (ECDSA) to the list of known hosts.  
  
  _ _ | _ _ | _ )  
 _ | ( _ /   Amazon Linux 2 AMI  
---| \ _ _ | _ _ |  
  
https://aws.amazon.com/amazon-linux-2/  
[ec2-user@ip-172-31-101-108 ~]$
```

Step 6: Verify and Access Resources

- Ensure that EC2 instance, S3 bucket, and VPC are successfully created.
- Access the EC2 instance using SSH or other relevant protocols.

```
drwxr-xr-x 17 user1 user1 4096 Nov  3 02:13 ..  
-rwxrwx-rw-  1 user1 user1 1700 Sep  1 00:29 blog01-key.pem  
user1@Ubuntu18b:~/Downloads$ chmod 400 ./blog01-key.pem  
user1@Ubuntu18b:~/Downloads$ ssh -i blog01-key.pem ubuntu@52.55.222.44  
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-1045-aws x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/advantage  
  
System information as of Wed Nov  3 06:17:43 UTC 2021  
  
System load:  0.0               Processes:            106  
Usage of /:   26.4% of 7.69GB   Users logged in:     0  
Memory usage: 23%              IPv4 address for eth0: 172.11.0.200  
Swap usage:   0%  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check  
Internet connection or proxy settings  
  
Last login: Tue Nov  2 13:46:09 2021 from 176.77.177.137  
ubuntu@ip-172-11-0-200:~$
```

Conclusion:

Thus, we completed hands-on experience in implementing Infrastructure as a Service (IaaS) using Amazon Web Services (AWS)